

The application of Braille on equipment and appliances is strictly governed by the ISO 17049:2013 standard. To ensure tactile readability, this standard dictates precise dimensional parameters: a dot base diameter of 1.0 to 1.7 mm (specifically 1.5 mm to 1.6 mm in many regional adoptions), a horizontal dot spacing of 2.0 to 2.8 mm, and a dot height of 0.3 to 0.7 mm. The biomechanics of tactile reading depend entirely on these precise dimensions, as the human fingertip must cover multiple dots simultaneously to recognize a cell pattern instantly. Fused Deposition Modeling (FDM) 3D printers using standard 0.4 mm nozzles struggle immensely to resolve a 1.5 mm domed dot reliably without severe stringing or layer-line artifacts that degrade tactile readability.

While "macro-scaling" the physical cell mathematically solves the mechanical FDM tolerance risk, it introduces a severe factual and logical error: a macro-scaled Braille cell violates the strict ISO 17049 dimensional parameters and actively destroys tactile readability, rendering the device useless for an actual Braille reader. If the distance between corresponding dots in adjacent cells exceeds the standard 6.1 mm to 7.6 mm, the user cannot read the display fluidly. Claiming to build a Braille display while intentionally violating the foundational standard that defines Braille geometry is a critical technical flaw that must be addressed. A catastrophic academic and architectural misalignment exists. The presentation is explicitly designated for the course "PBCST504," which is the KTU Microcontrollers course (Semester 5, 2024 Scheme). The official university syllabus for this course explicitly focuses on the ARM Cortex-M architecture—specifically the Cortex-M23 and Cortex-M33 processors—and mandates the use of STM32 microcontrollers (such as the STM32U575) for peripheral programming, TrustZone implementation, and FreeRTOS deployment.

However, Slide 11 states that the project utilizes an "ESP32-WROOM-32" development board. The ESP32 is based on the Tensilica Xtensa Dual-Core 32-bit LX6 microprocessor architecture, not the ARM Cortex-M architecture. Using an Espressif Xtensa-based microcontroller for a project assigned under a curriculum strictly dedicated to ARM Cortex-M and STM32 microcontrollers represents a fundamental failure to meet the core academic requirements of the course.

Second, the slide demonstrates a technical terminology error regarding the hardware itself. The text specifies the use of an "ESP32-WROOM-32 development board". The ESP32-WROOM-32 is a surface-mount module consisting of the SoC, flash memory, and an antenna. It is not a development board. The development board that houses this module, providing the USB-to-UART bridge, voltage regulation, and pin headers, is officially designated as the ESP32-DevKitC. Conflating a bare module with a populated development board is an elementary hardware engineering mistake.

Third, the electrical power architecture described in the slide requires immediate verification. The text states: "Six positional SG90-class micro servos drive eccentric cams," and "A 1,000 μ F capacitor is proposed across the servo power rail to help reduce transient voltage dips." The SG90 micro servo is a standard 9-gram actuator that operates via Pulse Width Modulation (PWM) at a 50Hz frequency, requiring 1ms to 2ms duty cycles. While the ESP32's LEDC peripheral features 16 independent hardware PWM channels capable of driving these signal lines simultaneously, the power draw is a major concern. An SG90 servo typically draws between 100mA and 250mA during standard movement, but stall currents can spike up to

700mA per servo. If all six servos move simultaneously or encounter mechanical resistance (e.g., binding against the FDM printed cams), the total current draw on the 5V rail could surge to between 1.5A and 4.2A. While a 1,000 μ F bulk decoupling capacitor will smooth short, high-frequency voltage transients and prevent the ESP32 from resetting (brownout) due to common ground loops, it is mathematically incapable of sustaining continuous high-current stall loads. The text critically omits any specification regarding the amperage rating of the external 5V power supply required to drive this load.”Basic Block Diagram.” A review of the slide reveals a critical omission: there is no diagram present. The slide contains only the title text and is otherwise completely blank. In a technical hardware presentation, the block diagram is essential for illustrating the interfacing between the microcontroller, the power distribution network, the PWM signal routing, and the electromechanical actuators. Submitting or presenting a slide deck with a missing primary visual graphic demonstrates a severe lack of quality assurance. . Ranked Top 5 Most Serious Errors

Syllabus/Architecture Mismatch (Slide 11): The presentation proposes using an ESP32 (Tensilica Xtensa architecture) for a project in a KTU course (PBCST504) that strictly mandates ARM Cortex-M and STM32 microcontrollers. This is a fatal academic failure that will result in immediate penalization during an engineering review.

Missing Block Diagram (Slide 12): Submitting an entirely blank slide designated for the core system architecture demonstrates a severe lack of quality control and will cause a critical stoppage during a live presentation.

ISO 17049 Dimension Violation (Slide 9): ”Macro-scaling” the Braille cell mathematically solves FDM printing tolerances but destroys the physiological capability of the fingertip to read the tactile data. The text must acknowledge this as a non-standard mechanical proof-of-concept rather than a functional Braille interface.

Hardware Terminology Conflation (Slide 11): Referring to the ”ESP32-WROOM-32” as a development board instead of a bare surface-mount module is a rudimentary hardware engineering error that undermines technical credibility. I have fact-checked the statements you provided against official technical documentation, standards, and university curriculum guidelines. Here is the verification for each point:

1. ISO 17049:2013 and Braille Dimensions

The statements regarding the Braille dimensions are accurate. The ISO 17049:2013 standard mandates strict dimensional parameters to ensure tactile readability, explicitly dictating a dot base diameter between 1.0 and 1.7 mm and a dot height between 0.3 and 0.7 mm. Regional accessibility guidelines that adopt these principles specify standard base diameters of 1.5 to 1.6 mm, dot spacing of 2.3 to 2.5 mm, and adjacent cell corresponding dot spacing of 6.1 to 7.6 mm. Therefore, intentionally ”macro-scaling” a Braille cell to bypass 3D printing tolerances would fundamentally violate these standardized metrics, rendering the display functionally useless for a tactile reader.

2. Academic Misalignment (KTU PBCST504 vs. ESP32)

The claim regarding the academic mismatch is factually correct. The APJ Abdul Kalam Technological University (KTU) 2024 Scheme defines course ”PBCST504” as ”Microcontrollers”

for Semester 5. The official syllabus is strictly dedicated to ARM Cortex-M architecture (specifically Cortex-M23 and M33 processors) and mandates the use of STM32 microcontrollers, such as the STM32U575. Conversely, the ESP32 operates on a Tensilica Xtensa dual-core architecture. Using an ESP32 for this specific project is a direct violation of the course's core architectural requirements.

3. Hardware Terminology (WROOM-32 vs. DevKitC)

The hardware terminology correction is highly accurate. The ESP32-WROOM-32 is a bare surface-mount module that contains the system-on-chip, memory, and a PCB antenna. It requires a carrier board to provide the USB-to-UART bridge, voltage regulation, and pinout headers for prototyping. The official Espressif development board that houses this module is the ESP32-DevKitC. Conflating the two is a technical terminology error.

4. Power Architecture and SG90 Servos

The technical assessment of the power architecture and current draw is accurate. The ESP32 features 16 independent hardware PWM channels (via its LEDC peripheral) capable of precisely driving multiple servos. However, while an SG90 servo draws 100mA to 250mA during standard movement, its stall current can spike from 360mA up to approximately 700mA. If six servos move simultaneously or stall, the combined current draw on the 5V rail can easily surge between 2.1A and 4.2A. A 1,000 μ F capacitor is meant to smooth transient dips; it cannot supply sustained high current. A dedicated external 5V power supply rated for the total stall current is mandatory to prevent system failure.

5. Missing Block Diagram

The critique regarding the missing visual element is logically and professionally sound. In hardware engineering presentations, a block diagram is a mandatory requirement to illustrate power distribution, signal routing, and component interfacing. Presenting a slide with only a title and no corresponding graphic is a critical quality assurance failure.